

Digital Hardware Implementation of a Sensorless Speed Controller for PMSM Drives Using HF Signal Injection and EKF

Hoang Than^{1,2,a}, Ying-Shieh Kung^{1,b} and Liang-Chiao Huang^{3,c}

¹Department of Electrical Engineering, Southern Taiwan University of Science and Technology,

No.1 Nan-Tai Street, Yong-Kang District, Tainan City 710, Taiwan, ROC

²Department of Electronics and Telecommunications, Hue Industrial College,

70 Nguyen Hue Street, Hue City, Vietnam

³Green Energy and Environment Research Laboratories, Industrial Technology Research Institute, Hsinchu 310, Taiwan, ROC

^ahthan@hueic.edu.vn, ^bkung@mail.stust.edu.tw, ^cliang.qiao@itri.org.tw

Abstract

Among the sensorless PMSM speed control algorithms, the non-model based approach, such as high frequency (HF) signal injection (HFSI), is usually taken as to estimate the rotor position of PMSM at standstill and very low speed condition, but the model based approach, such as extended Kalman filter (EKF) is suitable to the low/medium/high speed condition for PMSM drive. Therefore, to obtain a wide-speed range control from a low speed to a medium/high speed region, combining the aforementioned two approaches - HF signal injection and EKF in sensorless PMSM drive is necessary. In this paper, firstly, the mathematical modeling of PMSM and the rotor estimation algorithms of HFSI and EKF are derived. Secondly, a very high speed IC hardware description language (VHDL) is presented to describe the behavior of the adopted estimation algorithm. Thirdly, to verify the correctness of the designed VHDL code of the adopted estimation algorithm, based on electronic design automation (EDA) simulator link, a co-simulation work is constructed by Simulink and ModelSim which the input stimuli and output responses are run in Simulink and the estimation approach and controller are performed in ModelSim. Finally, some simulation results validate the effectiveness of the proposed sensorless PMSM speed control system.

Keywords: Sensorless speed control, PMSM, High frequency signal injection, Extended Kalman filter, ModelSim/Simulink co-simulation.

Introduction

Conventional PMSM control needs a position sensor or an optical encoder to measure the rotor position and speed; then feedback it to the controller for ensuring the precision current vector control and speed control. Such position sensor has some disadvantages such as drive cost, machine size, reliability and noise immunity. Therefore, a sensorless control without position sensor for PMSM drive becomes a popular research issue in literatures [1-6]. In those researches, there are the back electromotive force (Back EMF) [1], the sliding mode observer (SMO) [2], the extended Kalman filter (EKF) [3], the high-frequency signal injection (HFSI) [4-6], etc. The HFSI is insensitive to PMSM parameters and it can estimate the rotor position of PMSM at the low-speed and the standstill condition. Meanwhile the EKF is suitable for estimating the rotor position of PMSM at the medium/high speed condition. So, to extend a

wide-speed range control from a low speed to a medium/high speed region, combine two method HFSI and EKF together is the target of this paper.

To do that, a co-simulation work by electronic design automation (EDA) simulator Link have been used to test the effectiveness of HDL code in the motor driver system [7]. The EDA Simulator Link provides a co-simulation interface between Matlab/Simulink [8] and HDL simulator-ModelSim [9]. Hence, EDA Simulator Link lets us use Matlab code and simulink models as a test-bench that generates stimuli for a HDL simulation and analyses the simulation results response. In this paper, the PMSM model, inverter, speed command and switches are performed by Matlab/Simulink and the SVPWM, Speed controller, current controller, coordinate transformation, HFSI and EKF described by VHDL code is executed in ModelSim. Some simulation results validate the effectiveness of the proposed sensorless PMSM speed control system using HFSI and EKF.

Sensorless Speed Control for PMSM drives

The architecture of the sensorless PMSM driver speed control based on extended Kalman filter (EKF) and high frequency signal injection (HFSI) estimation is shown in Fig. 1. The modelling of PMSM, EKF estimator and the HFSI estimator are described as follows:

A. Mathematical Model of PMSM

The mathematical model of a PMSM is described, in two-axis d - q synchronous rotating reference frame, as follows

$$\begin{bmatrix} v_d \\ v_q \end{bmatrix} = \begin{bmatrix} r_s + sL_d & -\omega_e L_q \\ \omega_e L_d & r_s + sL_q \end{bmatrix} \begin{bmatrix} i_d \\ i_q \end{bmatrix} + \begin{bmatrix} 0 \\ \omega_e \lambda_f \end{bmatrix} \quad (1)$$

where v_d, v_q are the d and q axis voltages; i_d, i_q are the d and q axis currents, r_s is the phase winding resistance; L_d, L_q are the d and q axis inductance; ω_e is the rotating speed of magnet flux; λ_f is the permanent magnet flux linkage.

The current loop control of PMSM drive in Fig.1 is based on a FOC (Field Oriented Control) approach. After decoupling, the motor's torque is proportional to i_q .

$$T_e = \frac{3N_p}{4} \lambda_f i_q \triangleq K_t i_q \quad (2)$$

Considering the mechanical load, the overall dynamic equation of PMSM drive system is obtained by

$$J_m \frac{d}{dt} \omega_r + B_m \omega_r = T_e - T_L \quad (3)$$

where T_e is the motor torque, N_p is pole pairs, K_t is torque constant, J_m is the inertial value, B_m is damping ratio, T_L is the external torque, ω_r is rotor speed.

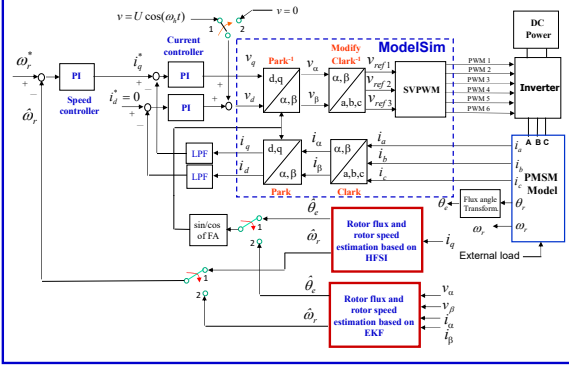


Fig. 1. The diagram of a sensorless control for PMSM driver with HFSI and EKF-based estimation

B. High frequency signal injection (HFSI) estimator

When PMSM operates in the low speed or near standstill condition, the ω_e approaches to zero; therefore, the cross coupling item and back EMF can be ignored. In addition, when a high frequency signal (ω_h) with $\omega_h L_q \gg r_s$ is injected, the (1) can be further reduced as

$$v_d = L_d \frac{di_d}{dt} \quad (4)$$

$$v_q = L_q \frac{di_q}{dt} \quad (5)$$

Before applying the HFSI, three reference frames are defined and shown in Fig. 2. The α - β axis is the stator reference coordinate; d - q axis is the actual rotating synchronous reference coordinate; $\hat{d}$ - $\hat{q}$ is the estimated rotating synchronous reference coordinate. The angle between d - q axis and α - β axis is θ_e , and between $\hat{d}$ - $\hat{q}$ axis and α - β axis is $\hat{\theta}_e$. Further, the estimated error is defined as

$$\Delta\theta_e = \theta_e - \hat{\theta}_e \quad (6)$$

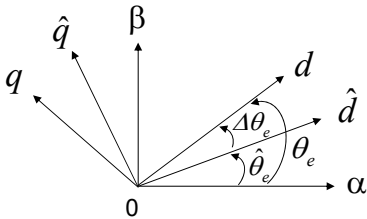


Fig. 2 The relationship among three reference frames

If a high frequency voltage is injected to $\hat{d}$ axis, the high frequency voltage equation in $\hat{d}$ - $\hat{q}$ axis is presented as follows

$$\hat{v}_{dh} = U_{mh} \cos(\omega_h t) \quad (7)$$

$$\hat{v}_{qh} = 0 \quad (8)$$

where $\hat{u}_{dh}, \hat{u}_{qh}$ are high frequency voltages in $\hat{d}$ - $\hat{q}$ axis; U_{mh} is the voltage magnitude; ω_h is the high frequency value. If the (7)-(8) are transformed from $\hat{d}$ - $\hat{q}$ axis to d - q axis, the high frequency voltage in d - q axis are respectively derived as follows

$$v_{dh} = U_{mh} \cos(\omega_h t) \cos \Delta\theta_e \quad (9)$$

$$v_{qh} = -U_{mh} \cos(\omega_h t) \sin \Delta\theta_e \quad (10)$$

where u_{dh}, u_{qh} are high frequency voltages in d - q axis. Further, substituting (9)-(10) into (4)-(5) and neglecting the inductance variation, the high frequency current response are

$$i_{dh} = \frac{U_{mh} \sin(\omega_h t) \cos \Delta\theta_e}{\omega_h L_d} \quad (11)$$

$$i_{qh} = -\frac{U_{mh} \sin(\omega_h t) \sin \Delta\theta_e}{\omega_h L_q} \quad (12)$$

where i_{dh}, i_{qh} are high frequency currents response in d - q axis. Continuously proceeding the coordinate transformation, high frequency currents response in $\hat{d}$ - $\hat{q}$ axis can be obtained as follows.

$$\hat{i}_{dh} = \frac{U_{mh}}{2\omega_h L_q L_d} ((L_q + L_d) + (L_q - L_d) \cos(2\Delta\theta_e)) \sin(\omega_h t) \quad (13)$$

$$\hat{i}_{qh} = \frac{U_{mh} (L_q - L_d)}{2\omega_h L_q L_d} \sin(2\Delta\theta_e) \sin(\omega_h t) \quad (14)$$

where $\hat{i}_{dh}, \hat{i}_{qh}$ are high frequency currents response in $\hat{d}$ - $\hat{q}$ axis, and from (13), it shows that $\hat{i}_{qh}$ includes $\Delta\theta_e$ information. Therefore, if we can control the $\Delta\theta_e$ to be zero, the rotor position can be estimated. To reach this goal, firstly, let $\hat{i}_{qh}$ multiplies $\sin(\omega_h t)$ function and the formulation is become as follows.

$$\hat{i}_{qh} \sin(\omega_h t) = \frac{U_{mh} (L_q - L_d)}{4\omega_h L_q L_d} \sin(2\Delta\theta_e) [1 - 2\cos(2\omega_h t)] \quad (15)$$

From (15), it consists of a dc component and a harmonic item. If this output in (15) sends to a low pass filter, as follows,

$$f(\Delta\theta) = LPF(\hat{i}_{qh} \sin(\omega_h t)) = \frac{U_{mh} (L_q - L_d)}{4\omega_h L_q L_d} \sin(2\Delta\theta_e) \quad (16)$$

the harmonic item will be filtered and only the dc component is left. The LPF in (16) represents low pass filter. Further, if we assume that the $\Delta\theta_e$ is small enough in (16), the $\sin(2\Delta\theta_e) \approx 2\Delta\theta_e$. Thus, a PI controller is applied to tune $\Delta\theta_e$ to zero, the initial rotor position θ_e can be estimated. The signal modulation process in HFSI algorithm is shown in Fig. 3.

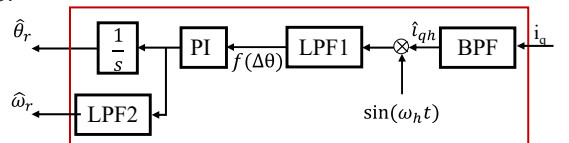


Fig. 3. The signal modulation process

C. Extended Kalman filter (EKF) estimator

From the mathematical models of PMSM, if it operates at the medium/high speed, the different inductance between the d and q axis is neglected and it can assume $L_d = L_q = L$. Under the assumption, the equation in (1) can be transformed into $\alpha - \beta$ axis, and presented as follows.

$$\begin{bmatrix} v_\alpha \\ v_\beta \end{bmatrix} = \begin{bmatrix} r_s + sL_s & 0 \\ 0 & r_s + sL_s \end{bmatrix} \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} + \omega_e \lambda_f \begin{bmatrix} -\sin \theta_e \\ \cos \theta_e \end{bmatrix} \quad (17)$$

Where $[v_\alpha \ v_\beta]^T$ is voltages on fixed coordinate; $[i_\alpha \ i_\beta]^T$ is current on fixed coordinate; θ_e is angular position at magnet flux; s is differential operator.

If we choose the $x(t) = [i_\alpha, i_\beta, \omega_e, \theta_e]^T$ as the state variable, $u(t) = [v_\alpha \ v_\beta]^T$ as the input and $y(t) = [i_\alpha \ i_\beta]$ as the output of the PMSM mathematical model, the motor stochastic nonlinear equation can be written by the following form

$$\dot{x}(t) = f[x(t)] + Bu(t) + \sigma(t) \quad (18)$$

$$y(t) = h[x(t)] + \mu(t) \quad (19)$$

and where $f[x(t)]$, B and $h[x(t)]$ are represented as follows,

$$f(x(t)) = \begin{bmatrix} -\frac{r_s i_\alpha}{L_s} + \frac{\omega_e \lambda_f}{L_s} \sin \theta_e \\ -\frac{r_s i_\beta}{L_s} - \frac{\omega_e \lambda_f}{L_s} \cos \theta_e \\ 0 \\ \omega_e \end{bmatrix} \quad (20)$$

$$B = \begin{bmatrix} \frac{1}{L_s} & 0 & 0 & 0 \\ 0 & \frac{1}{L_s} & 0 & 0 \end{bmatrix}^T \quad (21)$$

$$h = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \quad (22)$$

and the $\sigma(t)$ and $\mu(t)$ in (18)-(19) represent system noise and measurement noise which are zero-mean white Gaussian distribution with covariance $Q(t)$ and $R(t)$ respectively.

Further, the continuous model in (18)-(19) can be transformed to the equivalent discrete model and it is presented as follows.

$$x(n) = (I + FT_c)x(n-1) + BT_c \cdot u(n-1) + v(n-1) \quad (23)$$

$$y(n) = H \cdot x(n) + \xi(n-1) \quad (24)$$

where $t_n = nT_c$, $t_{n-1} = (n-1)T_c$ and T_c is the sampling time.

Further, the Jacobian matrix and transition matrix are

$$F(x(t)) = \frac{\partial f}{\partial x} \Big|_{x=x(t)} = \begin{bmatrix} -\frac{r_s}{L_s} & 0 & \frac{\lambda_f}{L_s} \sin \theta_e & \frac{\omega_e \lambda_f}{L_s} \cos \theta_e \\ 0 & -\frac{r_s}{L_s} & -\frac{\lambda_f}{L_s} \cos \theta_e & \frac{\omega_e \lambda_f}{L_s} \sin \theta_e \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad (25)$$

$$\Phi(t_n, t_{n-1}, x(t_{n-1})) \equiv I + FT_c = \begin{bmatrix} 1 - \frac{r_s T_c}{L_s} & 0 & \frac{\lambda_f T_c}{L_s} \sin \theta_e & \frac{\omega_e \lambda_f T_c}{L_s} \cos \theta_e \\ 0 & 1 - \frac{r_s T_c}{L_s} & -\frac{\lambda_f T_c}{L_s} \cos \theta_e & \frac{\omega_e \lambda_f T_c}{L_s} \sin \theta_e \\ 0 & 0 & 1 & 0 \\ 0 & 0 & T_c & 1 \end{bmatrix} \quad (26)$$

$$H(x(t)) = \frac{\partial h}{\partial x} \Big|_{x=x(t)} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \quad (27)$$

After the discrete model of the motor stochastic nonlinear equation is obtained, the EKF algorithm can be used to estimate the state value of $x(t) = [i_\alpha, i_\beta, \omega_e, \theta_e]^T$ in (18)-(19) from the measured input and output values. The EKF is an optimal estimator which searches the cost function $J = \sum_{n=1}^m E\{\tilde{x}^2(n)\}$ at the least square sense. The $\tilde{x}(n)$ is defined by $\tilde{x}(n) = \hat{x}(n) - x(n)$, which is the difference of estimation of state $\hat{x}(n)$ and system state $x(n)$. The EKF algorithm is described by the following two steps recursive equations:

Step 1: (Prediction step)

From (18) and using a simple rectangular integration

$$\hat{x}_{n|n-1} = \hat{x}_{n-1} + (f[\hat{x}_{n-1}] + B \cdot u_{n-1})T_c \quad (28)$$

The covariance is updated by

$$P_{n|n-1} = \Phi_{n-1} P_{n-1} \Phi_{n-1}^T + Q_d \quad (29)$$

Step 2: (Innovation step)

$$\hat{x}_n = \hat{x}_{n|n-1} + K_n (y_n - H\hat{x}_{n|n-1}) \quad (30)$$

$$P_n = P_{n|n-1} - K_n H P_{n|n-1} \quad (31)$$

The Kalman gain is calculated by

$$K_n = P_{n|n-1} H^T [H P_{n|n-1} H^T + R]^{-1} \quad (32)$$

Where $\hat{x}_{n|n-1}$, $P_{n|n-1}$ are respectively the estimated temporary state and temporary covariance from t_{n-1} to t_n . In addition, the initial values of Q_d , R and P_0 need to be chosen. As the results, the EKF algorithm in (28)-(32) can be carried out through the recursive calculation to estimate the state value $\hat{x}(n) = [\hat{i}_\alpha(n), \hat{i}_\beta(n), \hat{\omega}_e(n), \hat{\theta}_e(n)]^T$ at each sampling period, then the rotor speed can be derived by

$$\hat{\omega}_r(n) = \frac{\hat{\omega}_e(n)}{N_p} \quad (33)$$

Finally, the estimated rotor position $\hat{\theta}_e(n)$ will be fed back to current loop for vector control, and the estimated rotor speed $\hat{\omega}_r(n)$ will be fed back to speed loop for rotor speed control.

ModelSim/Simulink Co-Simulation of Sensorless PMSM driver with speed controller system

In Figure 1, the FOC, EKF estimator, HFSI estimator, speed controller and current controller described by VHDL code based on ModelSim. The PMSM, inverter and speed command are performed in Matlab/Simulink. The sensorless speed controller is executed in ModelSim and Simulink by four. The example of Simulink/ModelSim co-simulation for Sensorless PMSM drives only using EKF is shown Fig.4. It executes the function of the sensorless speed controller by three works. The work-1 performs the speed loop PI controller,

works-2 performs the FOC with the current PI controller, coordinate transformation and SVPWM generator and work-3 performs the EKF-based rotor position estimator. However, it has another work-4 for HFSI estimator (not show here). The sampling frequency in the current control loop and EKF-based estimator is designed with 16 kHz but in speed control loop is 2 kHz. All works of ModelSim is supplied by 50 MHz and 12.5 MHz pulse clocks.

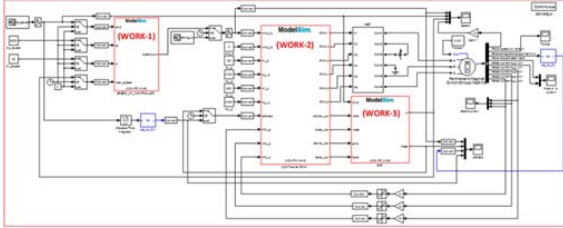


Fig. 4. The example of Simulink/ModelSim co-simulation for Sensorless PMSM drives (only using EKF)

In simulation, the designed PMSM parameters used in Fig. 4 are that pole pairs is 4, stator phase resistance is 1.3Ω , stator inductance is $L_q=6.3 \text{ mH}$, $L_d=4.3 \text{ mH}$, inertia is $J=0.000108 \text{ kg}\cdot\text{m}^2$ and friction factor is $F=0.0013 \text{ N}\cdot\text{m}\cdot\text{s}$. In the HFSI estimator design, a high frequency voltage (HFV) in d-axis is firstly injected with magnitude by 200 (0.1 unit) and 1kHz frequency cosin function. Secondly, the q-axis current can be acquired after detecting the three phase currents and through by Clark and Park transformation. Thirdly, the q-axis current will be sent to the signal modulation process in HFSI estimator for estimating the rotor position and rotor speed. In the design of the signal modulation process, the q-axis current is detected and sent into a 1kHz band-pass filter; then the output value multiplies with a 1kHz sinusoidal signal. Further, a first order low-pass filter with 100Hz cut-off frequency and a PID controller with $K_p=400$, $K_i=30$ and $K_d=0$ are cascaded with this signal. Via to the aforementioned signal modulation, the rotor speed can be estimated, and rotor position after integral the estimated rotor speed can be obtained. In the EKF estimator, the inputs are the currents and voltages in $\alpha - \beta$ axis, and the outputs are the estimated rotor position and rotor speed. Under the system, the simulation case is designed with that the HFSI estimator operates in the low speed ($\pm 60 \text{ rpm}$) then switches to EKF estimator for medium speed (300 to 400 rpm) after one second. The simulation result is shown in Fig. 5. The rotor position tracking in Fig. 5 (b) shows a very fast and accurate response, and the rotor speed tracking in Fig. 5 (a) show that it appears a little oscillation response in the low speed condition, but a better speed response in the medium speed condition.

Conclusion

This work has been successfully presented a sensorless speed control in PMSM drive using HFSI and EKF-based rotor position and speed estimation. This method is suitable to control the sensorless PMSM drive from the low speed to the medium/high speed with a fast tracking operation. In the future, this result will be confirmed on FPGA experiment.

Acknowledge

The financial support provided by Bureau of Energy, Ministry of Economics Affairs of Taiwan is gratefully acknowledged. The authors also gratefully acknowledge the

support of Green Energy and Environment Research Laboratories, Industrial Technology Research Institute, Taiwan.

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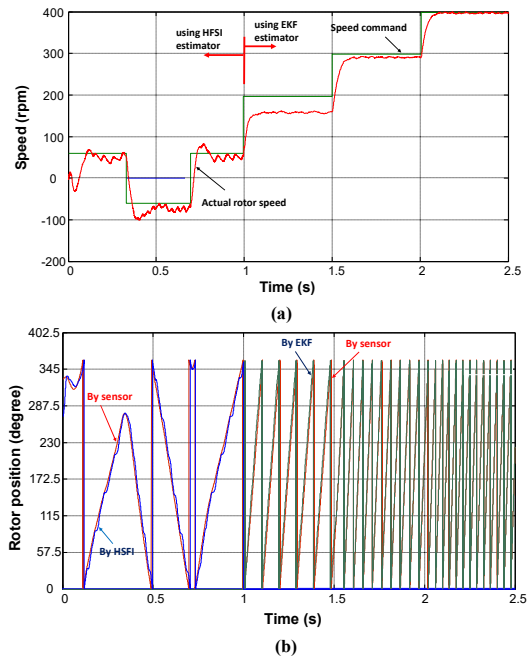


Fig. 5. Simulation result from the low speed to the medium speed
(a) rotor speed response (b) rotor position response